Regional variation in paleomagnetic polarity of the Matachewan dyke swarm related to the Kapuskasing Structural Zone, Ontario[1,2]

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The 2.45 Ga Matachewan dykes from the Abitibi Subprovince of the Canadian Shield yield a mean paleomagnetic pole of 42°N, 58°E (α_{95} = 3°; *N* (sites) = 36), which is a composite of new and previously published data. Domains of paleomagnetic polarity are defined: an area of dykes predominantly of reversed magnetization in the Abitibi Subprovince contrasts with an area of exclusively normal dykes to the north. The polarity domains are separated by faults related to the 1.95 Ga uplift and exposure of the lower crust in the Kapuskasing Structural Zone and therefore reflect Hudsonian age tectonics in the Archean Superior Province.

Les dykes de Matachewan de la sous-province de l'Abitibi, âgés de 2,45 Ga, fournissent un pôle paléomagnétique moyen de 42°N, 58°E (α_{95} = 3°; *N* (sites) = 36); ce pôle est composé de données nouvelles et anciennes déjà publiées. Les domaines de polarité paléomagnétique sont définis : ils font ressortir le contraste entre les dykes d'aimantation inversée, prédominants dans la sous-province de l'Abitibi, et les dykes de la région au nord, d'aimantation exclusivement normale. Les domaines de polarité sont séparés par des failles reliées au soulèvement il y a 1,95 Ga et à l'exposition de la croûte inférieure dans la zone structurale de Kapuskasing; ainsi ils reflètent une tectonique d'âge hudsonien dans la province du Lac Supérieur archéenne.

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Introduction

Numerous mafic dykes of Proterozoic age intruding Archean rocks of the Superior Province of Canada include the Matachewan dykes, which form an extensive swarm across the Abitibi and Opatica subprovinces of northern Ontario (Fig. 1). The Matachewan dyke swarm has been correlated with the Hearst dyke swarm to the west in the Wawa and Quetico subprovinces (Goodwin 1972; Watson 1980; Ernst and Halls 1984) (Fig. 1), and both swarms will be referred to as the Matachewan dykes. These dykes are generally vertical, strike north to north-northwest, and define a broad fan that becomes wider to the north. The dykes are thought to radiate from a southerly source related to failed rifting (Fahrig 1987).

Although much of the exposure is poor, the full extent of the swarm is outlined by aeromagnetic anomalies (Watson 1980; West 1988) (Fig. 2). The dykes continue southward beneath the Proterozoic Huronian Supergroup sediments (Aibangbee 1982) and northward beneath Paleozoic sediments of the Hudson Bay basin.

Early attempts at dating the Matachewan dykes were by K–Ar whole-rock analysis (Wanless *et al.* 1966; Leech 1966; Gates and Hurley 1973), which yielded a broad scatter of relatively young ages (2.49–1.33 Ga), and by Rb–Sr whole-rock analysis (Gates and Hurley 1973), which gave an age of 2.69 ± 0.09 Ga. The variation in the K–Ar dates probably results from argon loss subsequent to intrusion, and the Rb–Sr technique suffers from problems of wall-rock contamination, initial isotopic heterogeneity, and a lack of precision that is due to small variations in Rb/Sr (Hanes 1987). Recently, Heaman (1988) obtained a U–Pb age of 2.45 ± 0.002 Ga from analysis of primary magmatic zircons; this accurate and reliable method (Krogh *et al.* 1987) gives the best estimate of the time of emplacement of the Matachewan dyke swarm.

The dykes cross-cut the Archean age structural trends, thus postdating the Kenoran Orogeny, and are deflected across a major geological feature known as the Kapuskasing Structural Zone (KSZ) (Thurston *et al.* 1977) (Fig. 1), where discontinuous blocks of high-grade rocks that were once part of the lower crust are exposed (Percival and Card 1983, 1985). An abrupt change in metamorphic grade from granulite to greenschist occurs across the Ivanhoe Lake Cataclastic Zone (ILCZ), which defines the east side of the KSZ, but a more gradual change via amphibolite grade characterizes the west side (Percival 1983).

The western half of the Matachewan swarm has a change in trend that defines a broad Z pattern; the dykes generally trend northwest but swing to a more northerly trend where they cross the KSZ. Matachewan dykes appear to be absent from the granulite terranes within the KSZ but are present in the amphibolite terranes (Halls *et al.* 1985). Watson (1980) argued that the curvature in the dykes was a primary feature resulting from the regional stress field during intrusion and that deflection of the Matachewan dykes across the KSZ was due to sinistral strike slip along this zone as the dykes were emplaced. Percival and Card (1985) proposed east-southeast-directed thrusting along the ILCZ as an alternative explanation.

The Matachewan dykes have been the subject of earlier paleomagnetic studies (e.g., Strangway 1964; Fahrig *et al.* 1965; Pesonen 1973; Irving and Naldrett 1977) (see Fig. 1) and more recent studies as part of the Canadian Lithoprobe research program on the KSZ (Halls and Palmer 1988; Bates and Halls 1988). The results presented here represent part of an extensive paleomagnetic investigation of the Matachewan dykes in the vicinity of the KSZ.

Geological background

Structure

The majority of the paleomagnetic sites in this study come from Rocky Island Lake, situated in the Abitibi Subprovince of northern Ontario (Fig. 1) at latitude 46°55′N, longitude 83°05′W. The lake is a reservoir, and at normal water levels outcrop is limited. However, at the time of the survey in 1987, exposure was excellent because the water level had been lowered by about 5 m. Rocky Island Lake has an elongate form striking east-northeast and cuts across the swarm of Matache-

[1]Publication 121 of Lithoprobe Phase II.

[2]Publication 11 of International Geological Correlation Programme (IGCP) Project 257 on Precambrian Dyke Swarms.

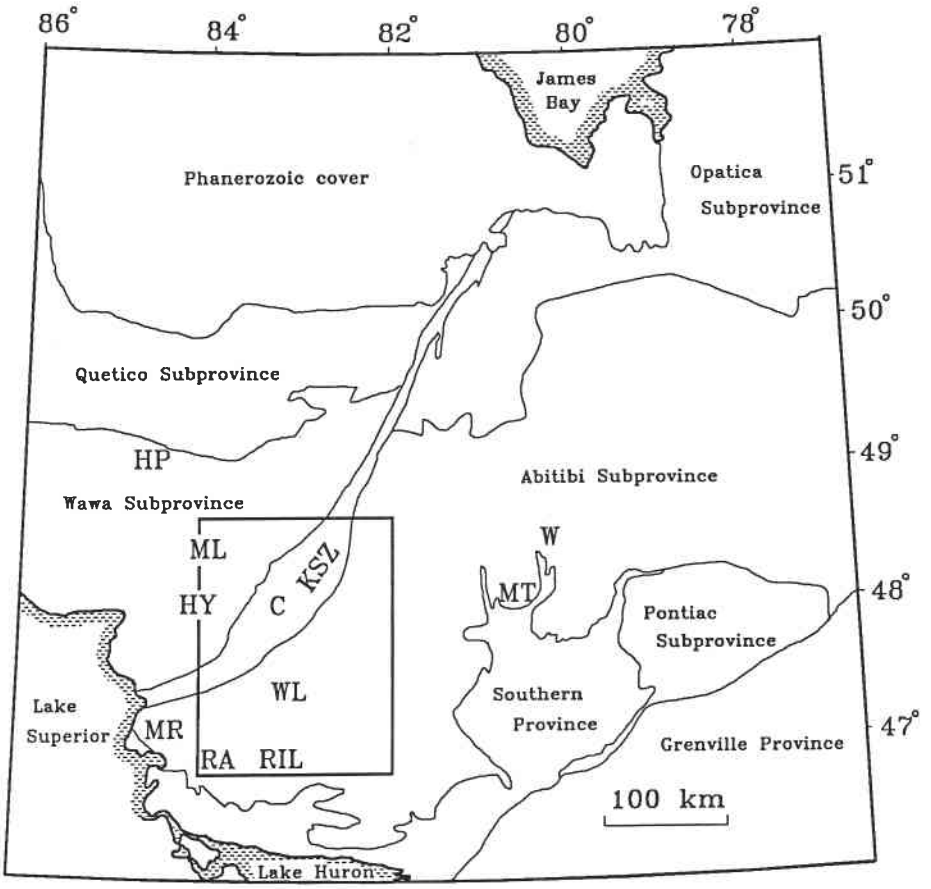


FIG. 1. Location of sites where Matachewan dykes have been sampled in this and previous studies. C, Chapleau; HP, Hornepayne; HY, Highway 101; KSZ, Kapuskasing Structural Zone; ML, Missinaibi Lake; MR, Montreal River; MT, Matachewan; RA, Ranger Lake; RIL, Rocky Island Lake; W, Wavell; WL, Wenebegon Lake. The box outlines the region of Fig. 2.

wan dykes (Fig. 3). The country rocks consist of uniform granitic gneiss of greenschist metamorphic grade (Percival and Card 1985). The dykes are heavily jointed and erode easily (compared with the host rocks), forming linear depressions. Most dykes are subvertical, with a bias towards northeast dips (≥85°), and strike about 330° with little variation, so that cross-cutting relations between the dykes are seldom observed.

The dykes range up to 60 m wide, have a mean width of 12 m, and are commonly associated with thin, parallel offsets and apophyses. Where the dyke swarm is relatively dense along two 0.5 km traverses, A and B (Fig. 3), estimates of crustal extension caused by dyke emplacement are 36 and 38%, respectively, but across the 35 km width of the swarm from Highway 129 to the eastern end of the lake the extension is only about 1.5%.

Further paleomagnetic sites were sampled at Wenebegon Lake, which lies 100 km north of Rocky Island Lake (Figs. 1, 4) and is also in the Abitibi Subprovince, at latitude 47°25′N, longitude 83°05′W. A number of vertical dykes, striking 350°, outcrop on the lake shore and islands, where country rocks are granitic, felsic, and mafic gneiss. On the basis of the aeromagnetic data, the dykes represent a sector of the swarm immediately to the east of those sampled at Rocky Island Lake.

Petrography

The Matachewan dykes are quartz tholeiites containing calcic plagioclase, pyroxene, amphibole, quartz, iron oxides, and sulphides, and their most distinguishing feature is the common occurrence of large, green, anorthositic plagioclase phenocrysts, up to 10 cm across (Condie *et al.* 1987), which are believed to have formed in shallow crustal magma chambers prior to dyke emplacement (Phinney *et al.* 1988). Phenocrysts are common in the dykes at Rocky Island and Wenebegon lakes and appear to have accumulated on the southeast side of constrictions in apophyses at site RL7 (Fig. 5). This phenomenon may be an indicator of a component of magma flow during dyke emplacement toward the northwest, consistent with the southerly source postulated by Fahrig (1987).

The dykes are green in the field, usually a sign that they have been hydrously altered. Thin sections reveal the typical suite of minerals described by Condie *et al.* (1987). However, plagioclase tends to be very dusty in appearance, some samples contain patches of fine clay minerals, and chlorite (which causes the green colour) is abundant in all samples. Occasionally, skeletal opaque minerals, almost completely replaced by chlorite, can be recognized. The alteration is observed both at the margins and within the main body of the dyke.

Paleomagnetism

Experimental procedure

We sampled 33 sites (46 dykes) from a 40 km traverse along Rocky Island Lake (Fig. 3) and five sites (five dykes) from Wenebegon Lake (Fig. 4). We took block samples of each dyke: the majority were taken from fine-grained chilled diabase at dyke margins to increase our chances of observing a stable remanence (Halls 1986). We oriented the blocks using both sun

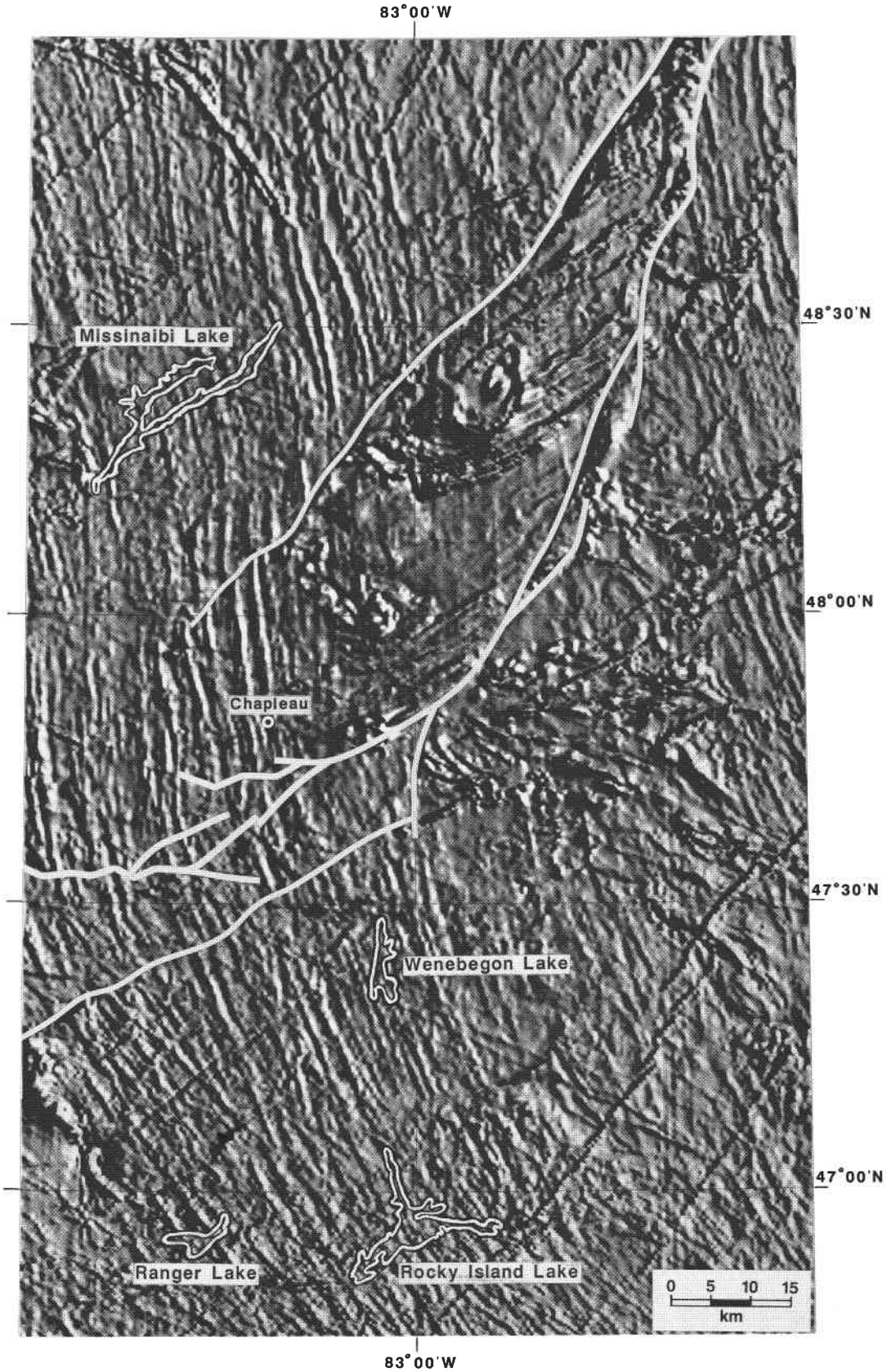


FIG. 2. Aeromagnetic shadowgram prepared by Urquhart-Dvorak Ltd. from Federal–Provincial surveys (800 m flight-line spacing, 300 m terrain clearance) digitized at 200 m intervals. After interpolation, pixels of 200 m side are used in the shadowgram, with a simulated light source from the east. Light intensity is proportional to the cosine of the angle between the normal to the field-intensity surface and the line joining the pixel to the light source. The position of the light source has been selected to emphasize the anomalies due to Matachewan dykes (aeromagnetic map courtesy of G. F. West). White lines are interpreted faults across which there appears to be a discontinuity in the dyke anomaly pattern.

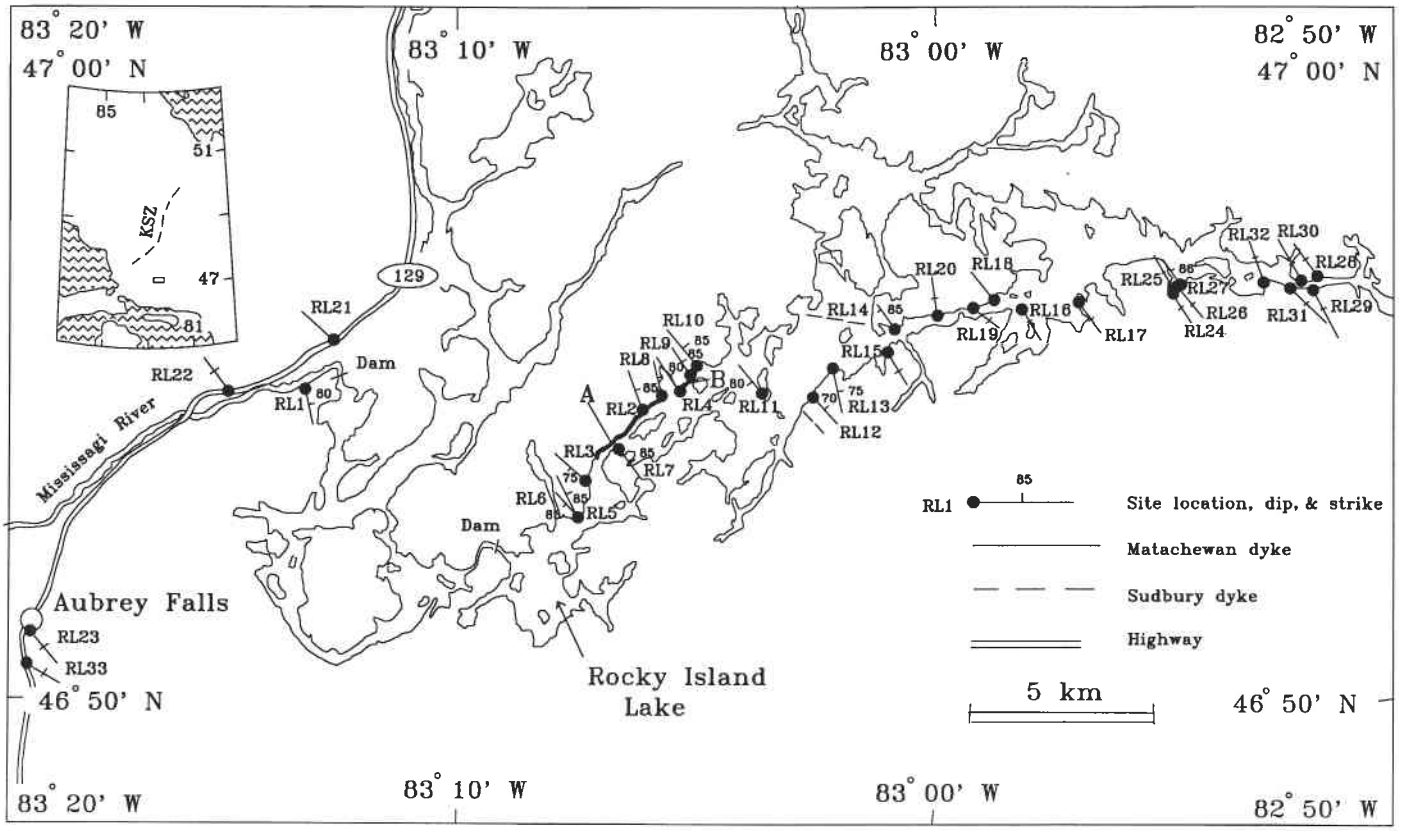


FIG. 3. Location of paleomagnetic sample sites of Matachewan dykes at Rocky Island Lake. Dyke widths and spacing were measured along traverses A and B.

and magnetic compasses, and where possible, we took up to eight blocks per dyke. Some closely located dykes share the same site number but are distinguished by the suffix W (west), C (centre), or E (east) as appropriate. Where dykes within a site carry the same remanence, they are considered branches of the same dyke and are represented by a single direction.

In the laboratory, blocks were drilled into individual cylindrical core specimens, 2.4 cm wide and 2.4 cm high. Each specimen was the subject of stepwise alternating-field (AF) demagnetization by a Schonstedt TD-1 instrument. Some thermal demagnetization was also carried out with a Schonstedt TSD-1, and remanent magnetization was measured on a modified DIGICO spinner magnetometer.

We analysed the resulting data using orthogonal vector plots (Zijderveld 1967), principal-component analysis (Kirschvink 1980), and remagnetization circles (Bailey and Halls 1984) in order to isolate individual components. If more than one specimen was analysed from a block sample, data were combined to give a sample mean before the site mean was calculated.

Results

The natural remanent magnetization (NRM) of samples has intensities that vary from less than 100 mA/m to over 10 000 mA/m, and a stable magnetization can be recovered from samples spanning the whole range of NRM intensities. The NRM is composed of one or two components. The most significant component occurs as one of two opposite polarities; a shallow, positively inclined (normal or N) north-northeast direction, and a shallow, negatively inclined (reversed or R) south-southwest direction. N and R are referred to collectively

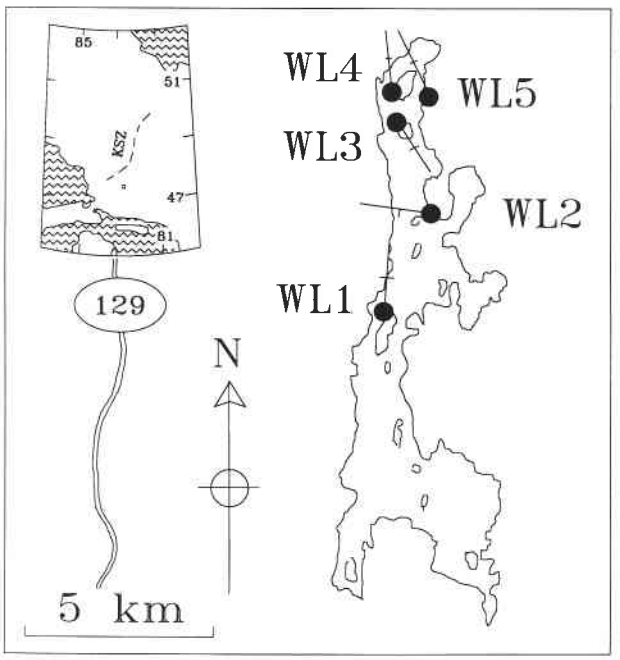


FIG. 4. Location of paleomagnetic sample sites of Matachewan dykes at Wenebegon Lake. Symbols as in Fig. 3.

as P and are defined by stable end points trending towards the origin (Fig. 6). P is found in samples from both chilled margins of some dykes and occurs at sites RL2E, RL8C, RL22, RL25, RL27, and WL1 as a single-component remanence, removed between 10 and 100 mT (the maximum available field) and with

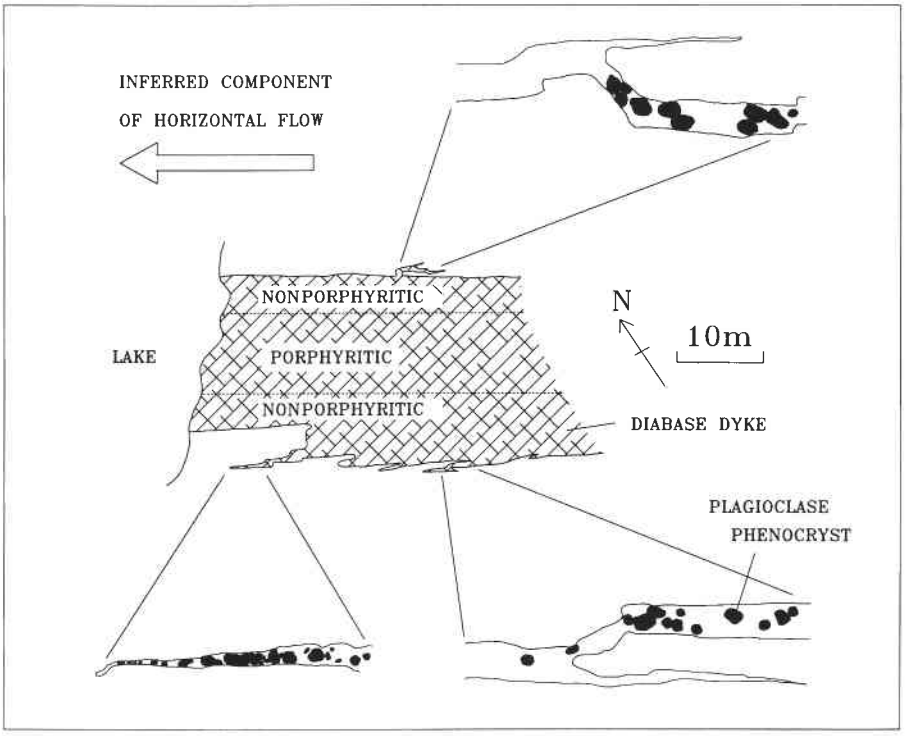


FIG. 5. Map of site RL7, showing plagioclase phenocrysts dammed on one side of constrictions in apophyses.

curves typical of remanence carried by single-domain magnetite (Dunlop 1983) (Fig. 7*a*).

Although examples of single-component R directions (e.g., A to D, Fig. 6) occur, the stable end point of R is usually achieved after 30–50 mT demagnetization. Most specimens also carry a soft remanence whose coercivity spectrum overlaps that of the R component (e.g., E and F, Fig. 6), so that the vector plots are initially curved. The overlapping may result if both N and R components are present in the same sample (e.g., G to I, Fig. 6), with N removed at lower coercivities. Alternatively, the overlap may be between R and the present Earth's field (PEF) ($\approx 352°/77°$; Garland 1979), and overprinting by later remagnetizations cannot be ruled out. N components are also resolved after removal of soft magnetizations (e.g., J, Fig. 6) but do occur as a single-component remanence as well (e.g., K and L, Fig. 6).

Thermal-demagnetization analyses show that P begins to be resolved as the Curie point of magnetite (580°C) is approached (Fig. 7*b*), but then the intensity of the magnetization falls rapidly and the specimen becomes unstable, so that a stable end point is rarely well defined. However, removal of a low-blocking-temperature component describes a great-circle path that converges on the site mean defined by AF demagnetization (Fig. 8). Thus AF is the preferred technique, but thermal data show that the high-coercivity component has a narrow blocking-temperature range consistent with it being a thermoremanent or chemical remanent magnetization carried by fine-grained, single-domain magnetite (Pullaiah *et al.* 1975).

Many specimens carry low-coercivity components (<10 mT) with distributed low-blocking-temperature ranges. These directions are poorly defined and scattered. The only significant-looking cluster of points falls close to the PEF direction, signifying a recent viscous origin.

The N site-mean directions often have slightly steeper inclinations than the R site-mean directions (Table 1; Fig. 9), although the difference is not significant at the 95% confidence limit. Even though stable end points were defined during demagnetization, further analysis was carried out to test whether the shallower R remanence is the result of our failing to completely remove a PEF overprint. If the R remanence contains two unresolved components, we would expect the R directions from any one site to lie along a plane between the two components. Although the sample-level data do not appear strung out at any site, we fitted best-fit planes to the R data from each site to test for a small but consistely oriented ellipticity to their distribution. However, the poles to these planes are spread out on a plane perpendicular to the mean R direction, thus indicating a lack of systematic streaking in the data. Therefore, the stable end points can be interpreted with confidence as single components.

The site-mean P directions for Rocky Island Lake and Wenebegon Lake are given in Table 1. Only those site means (indicated by ^{a} in Table 1) calculated from at least five independent samples and with $\alpha_{95} < 20°$ (Fig. 9) were used to calculate the overall N and R means (Table 2; Fig. 10). It is clear that the P direction is the component that is characteristic of the Matachewan swarm elsewhere (Figs. 1, 10; Table 2).

A minority of dykes failed to yield P directions, and some carry significant but anomalous high coercivity components (indicated by ^{b} in Table 1). These site means probably represent thermochemical overprints caused by post-Matachewan dyke intrusion events, possibly related to the alteration of the dykes, or are viscous overprints that have not been fully resolved. For example, a Sudbury dyke (age 1.25 Ga; Van Schmus 1975) was observed on several small islands to strike towards the Matachewan dyke at site RL14, which appears to have been totally remagnetized along the Sudbury dyke direction (270°/0°; Palmer *et al.* 1977). Other dykes that outcrop in this part of the Abitibi Subprovince include the 2.15 Ga Preissac and 1.15 Ga Abitibi swarms (Card *et al.* 1981). In addition, Hanes *et al.* (1988)

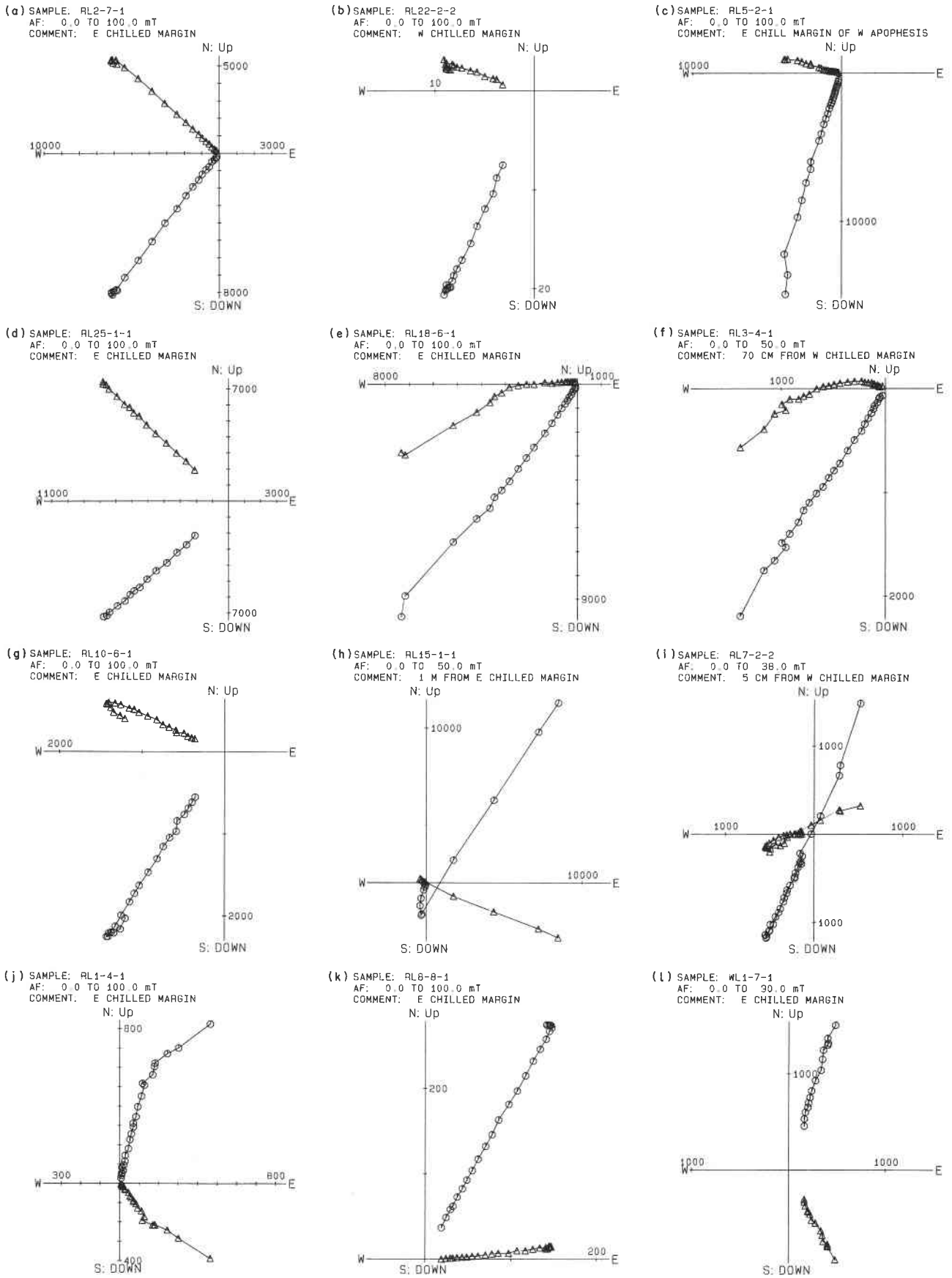


FIG. 6. Orthogonal vector plots of AF-demagnetization data (A/m $\times$ 10^{-3}) from the Matachewan dykes. Triangles: projection of data onto east–west vertical plane; circles: projection of data onto horizontal plane.

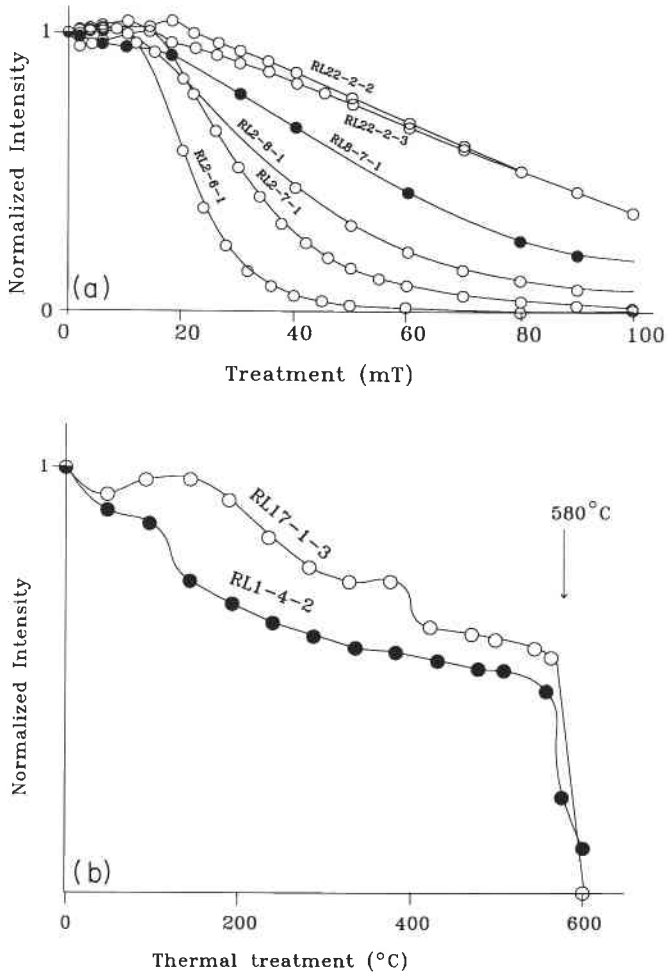


FIG. 7. NRM intensity-decay curves after (*a*) AF demagnetization and (*b*) thermal demagnetization on Matachewan dyke samples carrying a single-component remanence. Closed (open) symbols refer to data from normal (reversed) magnetization samples.

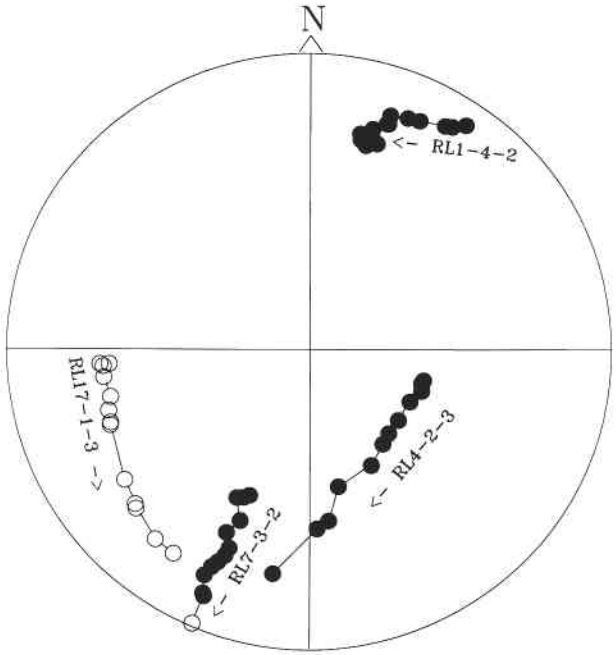


FIG. 8. Equal-angle stereoplots of thermal-demagnetization data. Open (closed) symbols refer to upward (downward) directions of magnetization.

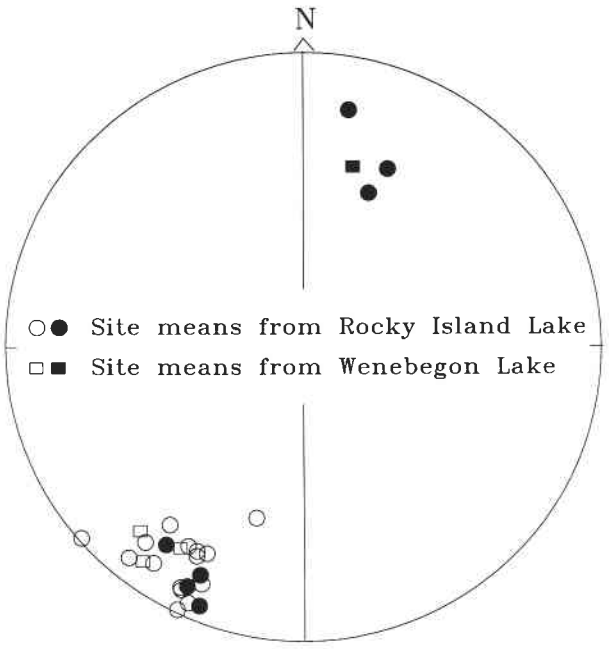


FIG. 9. Equal-angle stereoplot of site-mean N and R directions from sites used in the calculation of mean N and R directions. Open (closed) symbols refer to upward (downward) site means.

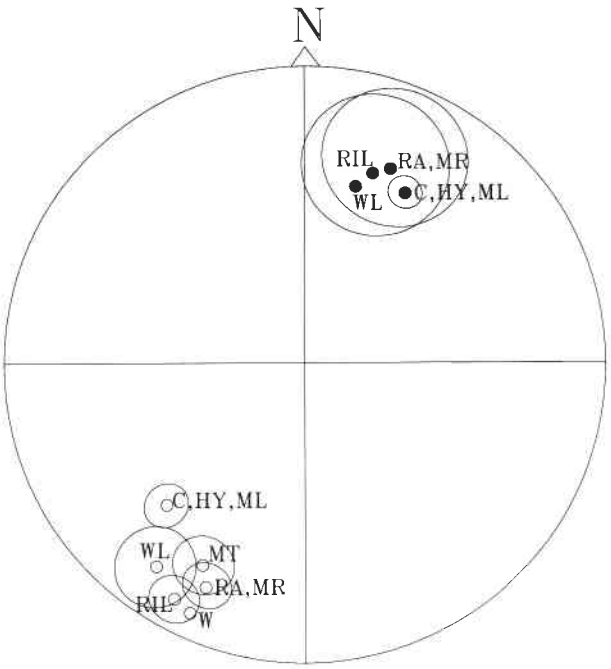


FIG. 10. Equal-angle stereoplot of mean N and R directions from this and previous studies. RIL, Rocky Island Lake; WL, Wenebegon Lake; W, Wavell (Pesonen 1973); MT, Matachewan (Irving and Naldrett 1977); RA, MR, Ranger Lake and Montreal River (Halls and Shaw 1988); C, HY, ML, Chapleau, Highway 101, and Missinaibi Lake (Halls and Palmer 1990). Open (closed) symbols refer to upward (downward) means. Circles of 95% confidence limit are shown; the WL normal and W reversed directions come from single sites and therefore do not have confidence limits.

postulated that the KSZ area underwent a low-temperature thermal event at 1.7 Ga. The Matachewan dykes are the oldest of the extensive swarms of the Superior Province, and the presence of later overprints is to be expected.

TABLE 1. Summary of Matachewan dyke paleomagnetic data

Site	Dyke width (m)	Dyke strike (°)	Dyke dip (°)	EP	RC	RS	D (°)	I (°)	k	α_{95} (°)	Polarity
					Rocky Island Lake						
RL1W	>12	333	90	5	—	1	204.1	9.5	54	10.5	R[a]
RL1E	24	345	89	5	—	2	10.9	10.5	250	4.7	N[a]
RL2W	23	327	90	6	—	—	22.5	30.0	270	3.5	N[a]
RL2E	3	280	83	3	—	—	213.3	−30.8	370	6.5	R
RL3	58	310	85	6	—	—	206.5	−13.1	720	2.5	R[a]
RL4	16	320	80	4	—	—	201.8	0.4	360	4.9	R
RL5	10	325	85	5	—	2	228.9	− 0.8	19	17.9	R[a]
RL6	33	140	87	4	—	1	205.7	5.9	37	15.3	R
RL7	21	328	85	5	—	—	214.7	12.0	16	19.5	R[a]
RL8W	17	165	85	3	2	—	223.7	3.5	9	34.0	R
RL8C	10	345	90	2	—	—	22.0	−4.0	110	23.7	N
RL8E	6	—	—	1	—	—	138.4	−69.3	—	—	—[b]
RL9	26	330	26	5	—	—	219.3	−4.7	63	9.7	R[a]
RL10	6	320	—	5	—	1	206.4	−5.0	21	16.9	R[a]
RL11	11	140	80	1	—	3	227.7	11.6	—	—	R
RL12	21	323	70	4	1	—	194.9	−28.4	28	17.0	R[a]
RL13	14	160	80	5	—	1	206.0	6.3	94	7.9	R[a]
RL14	17	339	85	5	—	—	273.7	19.2	330	4.2	—[b]
RL15	14	330	—	4	1	—	203.0	−7.7	90	9.4	R[a]
RL16	35	333	—	6	—	1	216.4	−16.2	40	10.8	R[a]
RL17	8	330	—	5	—	—	201.6	3.3	86	8.3	R[a]
RL18	9	270	—	5	—	3	218.8	−9.6	54	10.5	R[a]
RL19	17	315	90	3	—	1	221.8	−25.7	52	17.3	R
RL20	18	350	90	5	—	—	25.0	21.3	110	7.3	N[a]
RL21W	12	325	90	1	—	—	198.7	−4.1	—	—	R
RL21E	9	318	90	1	—	1	236.6	18.3	—	—	R
RL22	17	325	90	5	—	—	207.0	−5.4	200	5.5	R[a]
RL23	40	313	90	10	—	—	205.4	−1.0	64	6.1	R[a]
RL23X	0.5	316	90	2	—	—	202.6	−6.5	32	45.6	R
RL24	>2	326	90	2	—	—	204.9	4.0	30	47.5	R
RL25	4	331	90	5	—	—	209.8	−14.2	27	14.9	R[a]
RL26	22	320	90	5	—	—	214.5	−6.8	330	4.5	R[a]
RL27	>5	335	78	8	—	—	204.7	−14.7	64	7.0	R[a]
RL28	7	322	90	1	—	—	199.8	5.7	—	—	R
RL29	20	330	90	4	1	1	207.0	−14.1	41	14.0	R[a]
RL30	15	333	90	1	—	—	43.5	57.4	—	—	—[b]
RL31	12	312	90	5	—	1	44.2	61.3	360	4.0	—[b]
RL32	>12	334	—	—	—	3	—	—	—	—	—
RL33	12	308	90	5	—	2	204.2	−2.7	25	15.5	R[a]
					Wenebegon Lake						
WL1	6	4	90	5	—	—	15.7	24.9	93	8.0	N[a]
WL2	7	100	57	4	—	—	209.1	45.3	28	17.8	—[b]
WL3	10	338	90	4	2	—	221.2	−10.5	27	16.0	R[a]
WL4	16	356	90	5	—	—	211.4	−12.7	39	12.5	R[a]
WL5	15	340	90	4	1	—	216.7	−5.9	62	11.8	R[a]

NOTES: Sites that yield anomalous directions are RL8E, RL30, RL31, and WL2. The dyke at WL2 has a dip and strike different from those of the Matachewan swarm and may belong to a different swarm. The result from RL8E is based on a single block from the coarse-grained interior of an altered dyke. The remanence at sites RL30 and RL31 has not previously been reported from the Abitibi Subprovince and remains unexplained. Dyke X at site RL23 is a thin dykelet found within dyke RL23. EP, number of samples that yield stable end points; RC, number of other samples that yield remagnetization circles; RS, number of samples that have been rejected; D and I, declination and inclination of the remanence direction; k, precision parameter; α_{95}, radius of the 95% confidence circle about the mean site direction; N, normal; R, reversed.

[a]Site-mean values included in the calculation of overall mean values (at least five independent samples and with α_{95} of <20°).

[b]Sites that fail to yield a Matachewan direction.

Discussion

The dual-polarity P (i.e., N and R) remanence identified in this study is similar in direction and demagnetization characteristics to the remanence of the Matachewan dykes from previous studies (Fig. 10; Tables 2 and 3). Correction for geographic location allows the calculation of a paleomagnetic pole for all Matachewan dykes of the Abitibi Subprovince, which we obtained by combining data from this study with those of Irving and Naldrett (1977) and Halls and Shaw (1988) and with data from one site at Wavell, near Kirkland Lake, from Pesonen (1973). The data of Halls and Palmer (1990) come from the KSZ and the Wawa Subprovince and are not included. Using the

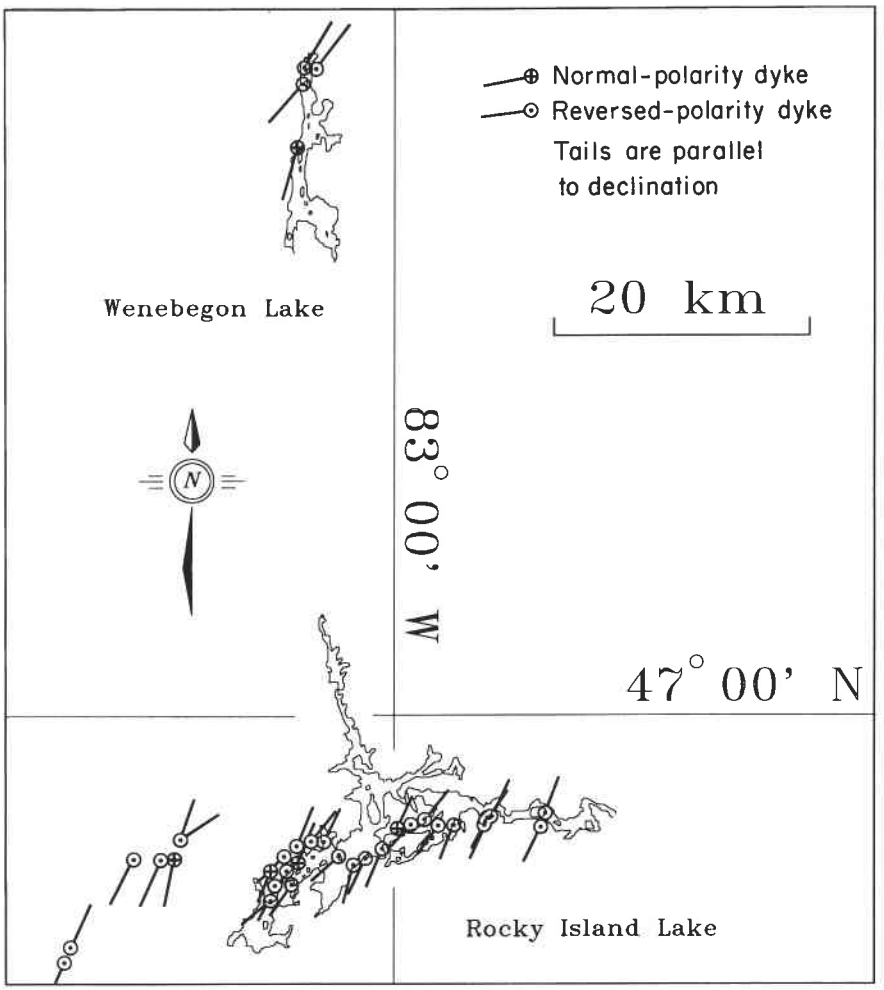


FIG. 11. Distribution of normal- and reversed-polarity dykes at Rocky Island and Wenebegon lakes. The strike of the paleomagnetic declination is shown for each dyke to demonstrate the lack of any significant relative rotations of individual dykes about vertical axes.

TABLE 2. Mean paleomagnetic remanence of parts of the Matachewan dyke swarm from this and previous studies

Location	Polarity	N	D (°)	I (°)	k	α_{95} (°)
RIL[a]	N	3	19.2	20.7	45	18.6
RIL[a]	R	19	209.2	−6.0	42	5.2
WL[a]	N	1	15.7	24.9	—	—
WL[a]	R	3	216.5	−9.7	190	9.1
W	R	1	205.6	−7.4	—	—
MT	R	10	207.0	−16.0	54	7.0
RA, MR	N	3	23.2	17.9	49	17.9
RA, MR	R	17	204.1	−11.6	52	5.4
C, HY, ML	N	37	30.1	22.4	66	2.9
C, HY, ML	R	19	224.2	−23.4	41	5.3

NOTES: RIL, Rocky Island Lake; WL, Wenebegon Lake; W, Wavell, Cook Township (Pesonen 1973); MT, Matachewan (Irving and Naldrett 1977); RA, MR, Ranger Lake and Montreal River (Halls and Shaw 1988); C, HY, ML, Chapleau, Highway 101, and Missinaibi Lake (Halls and Palmer 1990). The studies by Strangway (1964) and Fahrig *et al.* (1965) are not included because the AF values used (≤10 mT) did not completely resolve components. The data of Ernst and Halls (1984) are superseded by those in Halls and Palmer (1990), and only those dykes unaffected by Keweenawan rifting along Lake Superior have been included in the mean values from Halls and Shaw (1988). Locations of studies are shown in Fig. 1. N, number of site means.
[a]This study.

same site-selection criteria as for Table 1 and assuming an axial geocentric dipole model, we calculated the pole to be 42°N, 58°E (α_{95} = 3°; number of sites, N = 36; Table 3).

Pesonen (1973) found that the baked contact of Matachewan dykes also carried the same remanence as the dykes, but unfortunately the unbaked country rocks were paleomagnetically unstable, and thus the test was inconclusive. Irving and Naldrett (1977) described a baked-contact test with the Kamiskotia gabbro that suggested the N direction was primary, although they could not demonstrate conclusively the older age of the gabbro remanence. However, Schutts and Dunlop (1981) and Buchan *et al.* (1989) obtained positive baked-contact tests in separate studies of Matachewan dykes from the Kirkland Lake area: the Matachewan dykes clearly remagnetized the country rocks at their margins at or near the time of intrusion.

In the Missinaibi Lake area (Figs. 1, 2), Matachewan dykes carrying exclusively N directions cut, and therefore postdate, those dykes carrying exclusively R directions (Halls and Palmer 1988). This age relationship may hold true for the Rocky Island and Wenebegon Lake Matachewan dykes but cannot be proved because of a lack of data. There are only two examples of intersecting dykes in this study: at site RL8, a younger N dyke (RL8C) cuts an older dyke that failed to yield an interpretable result; and at site RL23 both dykes have R polarity.

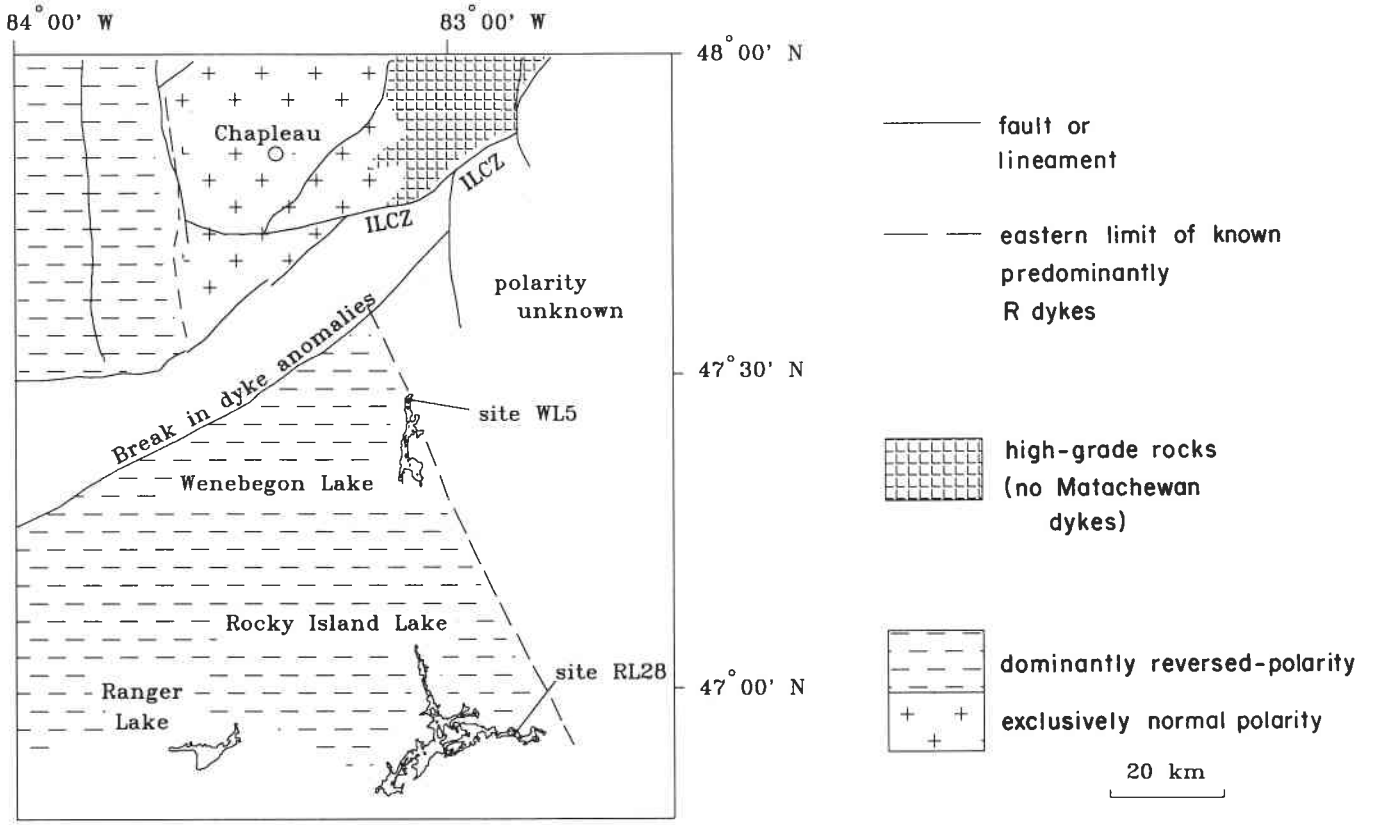


FIG. 12. Comparison of the polarity of the Matachewan dykes from north and south of the ILCZ, at the southern end of the KSZ. Sites WL5 and RL28 mark the eastern limit of the palaeomagnetic data of this study and hence the known easterly extent of the reversed-polarity zone. The discontinuity in the dyke aeromagnetic-anomaly pattern south of the ILCZ was recognized from Fig. 2, although it does not correspond to any known fault. No paleomagnetic data on the dykes are available for the area between the discontinuity and the ILCZ because of poor exposure.

TABLE 3. Paleomagnetic poles for the Matachewan dykes from this study and for the whole Abitibi Subprovince, including data from this and previous studies (Irving and Naldrett 1977; Halls and Shaw 1988; Pesonen 1973)

Location	Polarity	*N*	Lat. (°N)	Long. (°E)	α_{95} (°)
RIL	N	3	50	66	14
RIL	R	19	40	58	4
RIL	N and R	22	41	59	4
WL	N	1	53	70	—
WL	R	3	37	49	8
WL	N and R	4	42	53	13
ASP	N and R	36	42	58	3

NOTES: Poles were obtained from means of virtual geomagnetic poles (VGP) calculated for individual sites that pass the selection criteria (at least five independent samples and with α_{95} of <20°). Poles from both normal (N) and reversed (R) dykes, as well as the combination of both, are given for each study area for comparison. RIL, Rocky Island Lake; WL, Wenebegon Lake; ASP, Abitibi Subprovince. *N*, number of VGP's.

For samples that carry both N and R components, N is always carried in the lower coercivities and probably reflects a remagnetization partially overprinting a primary R component.

The baked-contact tests, cross-cutting relationships, high coercivities and unblocking temperatures, and overall consistency of the remanences of the Matachewan dykes strongly imply a primary intrusive or early hydrothermal origin of the P remanence. Thus the magnetization has an age of about 2.45 Ga, predating the development of the KSZ, which is believed to have been uplifted at about 1.95 Ga (Percival *et al.* 1988).

The characteristic directions obtained from the Matachewan dykes of the Abitibi Subprovince are similar to those from the vicinity of the KSZ (Fig. 10). There is no evidence for tectonic rotation that might be ascribed to westerly tilting of the crust in the KSZ (Percival and Card 1983, 1985). Matachewan dykes have not been identified within the highest grade (granulite) rocks of the KSZ and thus provide no information on tilting in a 30 km strip of the KSZ immediately west of the ILCZ.

The magnetic polarity of the Matachewan dykes undergoes a distinct change along the strike of the swarm. In the KSZ, a band of 25 dykes shows only N polarity (Halls and Palmer 1990). This contrasts with the Rocky Island and Wenebegon Lake areas to the south, where of the 38 dykes that give Matachewan directions, 33 have R polarity and only five have N polarity (Figs. 11, 12). The aeromagnetic map (Fig. 2) shows that the dykes in one polarity domain cannot be traced into the other but are discontinuous across the ILCZ and related faults. Three other less well-defined domain boundaries lie along the northwestern margin of the KSZ and cross Highway 101 west of Chapleau (Halls and Palmer 1990); in each case, the change in polarity also closely corresponds to previously known or inferred faults. Based on petrological differences between dykes within and outside the KSZ and on the observed cutting of R dykes by those of N polarity, Halls and Palmer (1990) propose that the domain boundaries are caused by vertical fault movements, perhaps of several kilometres, that have juxtaposed

R dykes against a younger N set that was intruded at a deeper crustal level. An alternative explanation is that the N dykes within the KSZ acquired their remanence upon uplift and cooling, but the cross-cutting relationships and close spatial relationship of N and R dykes outside the KSZ indicate that they were probably injected during normal and reverse polarity fields, respectively. Transcurrent faulting may be the cause of the polarity discontinuities along the KSZ boundary faults, but the dyke pattern cannot be matched across the ILCZ; the aeromagnetic map (Fig. 2) indicates relatively few Matachewan dykes east of either the N or R domain in Fig. 12, and from Rocky Island Lake westwards to Lake Superior R dykes continue to significantly outnumber those of N polarity (23 to 4; Halls and Shaw 1988), so that the N zone is absent from this sector of the dyke swarm.

Conclusions

This paper presents the first convincing documentation of a major paleomagnetic polarity discontinuity between Matachewan dykes lying within and outside the KSZ, a discontinuity that appears to be related to uplift along the ILCZ and related faults. The occurrence of polarity-domain boundaries in the Matachewan swarm is of great significance because they may constitute a practical means for mapping major faults in the Superior Province, especially those in gneissic terranes, where lithological and magnetic contrasts may be small.

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